Question	Answer	Marks
1	**Defining the problem**	
	A is the independent variable and *s* is the dependent variable **or** vary *A* and measure *s*	**1**
	keep *B* and *t* constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • pin / rod through hole • supported by a stand • sheet able to oscillate freely • at least one label from copper/sheet, hole, clamp, stand, rod, pin.	**1**
	drawn clamped rule(r) parallel to the direction of the oscillations (by eye) (to measure *s*)	**1**
	use rule(r) to measure lengths to determine *A* **and** *A* = length × breadth	**1**
	use of micrometer to measure *t*	**1**
	Method of Analysis	
	plot a graph of ln *s* against *A* or equivalent	**1**
	relationship valid if a straight line (with *y*-intercept = ln s_0) is produced	**1**
	$K = -\frac{\text{gradient}}{Bt}$ ($K = -\frac{1}{Bt \times \text{gradient}}$ for *A* against ln *s*)	**1**

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Question	Answer	Marks
1	**Additional detail including safety considerations**	**6**
	D1 use of cushion/sand box <u>in case sheet falls</u> **or** use gloves to protect hands from <u>cuts / sharp edges</u>	
	D2 keep (initial) distance between (copper) sheet and (poles of) magnet <u>constant</u> **or** keep (initial) distance between (copper) sheet and coil(s) <u>constant</u>	
	D3 keep s_0 <u>constant</u>	
	D4 method to ensure s_0 is constant, e.g. initially line up (corner of) plate with fiducial marker / vertical pin to keep s_0 constant	
	D5 method to determine *s* using video camera: • rule(r) in a position to measure *s* in the diagram • video camera shown in diagram or description of use of video camera • playback video recording by frame by frame / slow motion (to measure *s*)	
	D6 repeat measurements of *t* <u>in different positions</u> and <u>average</u> *t*	
	D7 measure *B*/magnetic flux density using a (calibrated) Hall probe	
	D8 additional detail on use of Hall probe, e.g. adjust probe until maximum value **or** measure *B* using Hall probe first in one direction and then in the opposite direction and average	
	D9 drawn method to create a magnetic field perpendicular to the area of the sheet, e.g. pair of magnets/horseshoe magnet/pair of coils connected to a (d.c.) supply	
	D10 repeat experiment for each *A* and average *s*	

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